Fundamentele programării

6_8. $T(n) = 2n \in \Theta(n)$

6_9. $T(n) = n \cdot 2n = 2n^2 \in \Theta(n^2)$

$$\sum_{i=0}^{n-1} \left(\sum_{j=0}^{n-1} 1 + \sum_{k=0}^{n-1} 1 \right) = \sum_{i=0}^{n-1} n+n = 2n \sum_{i=0}^{n-1} 1 = 2n \cdot n$$

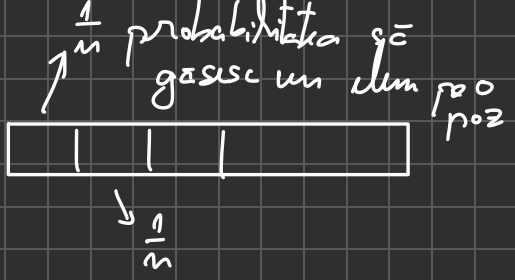
($\sum_{j=0}^{n-1} 1$: $\downarrow n$; $\sum_{k=0}^{n-1} 1$: $\downarrow n$)

6_11. $\Theta(n^2 \cdot \log_{10} n)$

BC

WC

AC $\sum_{i \in D} P(i) \cdot E(i)$ (prob · pașii pt a ajunge la el)




```
def search(lst, el):
    for x in lst
        if x == el
            return True
    return False
```

BC = elem pe prima poz $\Theta(1)$

WC = $el \notin lst$ $\Theta(n)$

$$AC = \underbrace{\sum_{i \in D}}_{\text{toate intrările pos}} P(i) E(i) = \overbrace{1}^{\text{el pe prima poz}} \cdot \frac{1}{n} + \overbrace{2}^{\text{el a doua poz}} \cdot \frac{1}{n} + 3 \cdot \frac{1}{n} + \dots + n \cdot \frac{1}{n} =$$

(2: $\downarrow$ 2 pași)

$$= \frac{1+2+3+\dots+n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2} \in \Theta(n)$$

OVERALL COMPLEXITY: O(n)

```
recursive_f_1
  if n <= 0
    return 1
  else
    return 1 + recursive_f_1(n-1)
```

f_1(5)
- f_1(4) ✗
  - f_1(3)
    - f_1(2)
      - f_1(1)
        - f_1(0)

$$T(n) = \begin{cases} 1 & \text{dacă } n <= n \\ 1 + T(n-1) & \text{altfel} \end{cases}$$ ✓

$T(n) = 1 + T(n-1)$

$T(n-1) = T(n-2) + 1$

$T(n-2) = T(n-3) + 1$

...

$T(1) = T(0) + 1$

$T(0) = 1$

$T(n) = \underbrace{1+1+1+\ldots+1}_{n+1} = n+1 \in \Theta(n)$

---

recursive_f_2

$$T(n) = \begin{cases} 1 \text{ dacă } n \leq 1 \\ 1 + T(n-5) \text{ altfel} \end{cases}$$

$T(n) = T(n-5) + 1$

$T(n-5) = T(n-10) + 1$

$\vdots$

$T(K) = 1$, $K \leq 1$

$T(n) = \underbrace{1+1+1+\ldots+1}_{n/5+1} \in \Theta(n)$

---

recursive_f_3

$$T(n) = \begin{cases} 1 \text{ dacă } n \leq 0 \\ 1 + T(n/2) \text{ altfel} \end{cases}$$

$T(n) = T(n/2) + 1$

$T(n/2) = T(n/4) + 1$

$T(n/4) = T(n/8) + 1$

...

$\Theta(\log_2 n)$

---

rekursiva funktion

$$T(n) = \begin{cases} 1 & \text{om } n \leq 0 \\ 2T(n-1) + 1 & \text{annars} \end{cases}$$

$T(n) = 2T(n-1) + 1$

$T(n-1) = 2T(n-2) + 1$

$T(n-2) = 2T(n-3) + 1$

⋮

$T(n) = 2T(n-1) + 1$

$= 2[2T(n-2)+1] + 1$

$= 2^2[2T(n-3)+1] + 2 + 1$

⋮

$= 2^K[2T(n-K)+1] + 2^{K-1} + 2^{K-2} + \ldots + 1$

$K = n$

$2^n T(0) + 2^{n-1} + \ldots + 1$

$= 2^{n+1} - 1 \in \Theta(2^n)$

def rucursive_fun_s

$$T(n) = \begin{cases} 1 \text{ dacă } n \leq 1 \\ n + T(n-1) \text{ altfel} \end{cases}$$

$T(n) = T(n-1) + n$

$T(n-1) = T(n-2) + n-1$

$T(n-2) = T(n-3) + n-2$

⋮

$T(2) = T(1) + 2$

$T(1) = T(0) + 1$

$T(0) = 1$

$T(n) = 1 + 1 + 2 + 3 + \ldots + n = 1 + \frac{n(n+1)}{2} \in \Theta(n^2)$

recursivă - § - 6

$$T(n) = \begin{cases} 1 & \text{dacă } n \leq 1 \\ 4T(n/2) + 1 & \text{altfel} \end{cases}$$

$T(n) = 4T(n/2) + 1$

$T(n/2) = 4T(n/4) + 1$

$T(n/4) = 4T(n/8) + 1$

$T(n) = 4T(n/2) + 1$

$= 4(4T(n/4) + 1) + 1$

$= 4^2 T(n/4) + 4 + 1$

$= 4^2 [4T(n/8) + 1] + 4 + 1$

$= 4^3 T(n/8) + 4^2 + 4 + 1$

...

$= 4^k T\left(\frac{n}{2^k}\right) + 4^{k-1} + \ldots + 4 + 1$

pt $n = 2^k$ $\quad T(n) = 4^k T(1) + 4^{k-1} + \ldots + 4 + 1 = \frac{4^{k+1} - 1}{3}$

$$x^n + x^{n-1} + x^{n-2} + \ldots + x + 1 = \frac{x^{n+1} - 1}{x - 1}$$

$\int_{-12}$

complexitatea timp : $O(n \log n)$

EXTRA-SPACE $\Theta(1)$
COMPLEXITY

$\int_{-13}$

COMPLEXITATEA TIMP : $T(n) = \begin{cases} 1 \text{ dacă } n \leq 1 \\ 2T(n/2)+1 \text{ altfel} \end{cases}$

$T(n) = 2T(n/2)+1$

$T(n/2) = 2T(n/4)+1$

$T(n/4) = 2T(n/8)+1$

⋮

$T(n) = 2T(n/2)+1$

$= 2[2T(n/4)+1]+1$

$= 2^2 T(n/4) + 2 + 1$

$= 2^2[2T(n/8)+1] + 2 + 1$

$= 2^3 T(n/8) + 2^2 + 2 + 1$

⋮

$= 2^k T(n/2^k) + 2^{k-1} + \dots + 1$

$n = 2^k$

$2^k \underbrace{T(1)}_{=1} + 2^{k-1} + \dots 1$

$= 2^k + 2^{k-1} + \dots + 1 =$

$= 2^{k+1} - 1 = 2 \cdot \underbrace{2^k}_{n} - 1 = 2n - 1 \in \Theta(n)$

SPACE COMPLEXITY

$$T(n) = \begin{cases} 1 & \text{if } n \leq 1 \\ 2T(n/2) + n & \end{cases}$$

↓ slicing ⇒ list of copies

$T(n) = 2T(n/2) + n$

$T(n/2) = 2T(n/4) + n/2$

$T(n/4) = 2T(n/8) + n/4$

⋮

$T(n) = 2T(n/2) + n$

$= 2[2T(n/4) + n/2] + n$

$= 2^2 T(n/4) + n + n$

$= 2^2[2T(n/8) + n/4] + n + n$

$= 2^3 T(n/8) + n + n + n$

⋮

$2^k T(n/2^k) + k \cdot n$

$n = 2^k \Rightarrow k = \log_2 n$

$\underbrace{2^k}_{n} T(1) + k \cdot n$

$n + k \cdot n =$

$= n + \log_2 n \cdot n$

$\in \Theta(n \log n)$